

P. Shubham Parashar

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THEORETICAL PHYSICIST

RESEARCH FOCUS

Theoretical condensed-matter physicist working on quantum transport, kinetic theory, electron hydrodynamics, magnetotransport, spatially resolved response, and quantum Hall systems. I ask when a spatial transport measurement can distinguish microscopic kinetic dynamics from an effective fluid description, and where low-moment hydrodynamic closures cease to be complete.

SELECTED PUBLICATIONS AND CURRENT MANUSCRIPTS

CURRENT MANUSCRIPTS

Complete kinematic null for local kinetic dissipation in a sixfold-driven electron fluid. P. Shubham Parashar.
Under review at *Physica B* (2026).

Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport. P. Shubham Parashar.
Under review at *Physical Review B* (2026). [arXiv:2607.22730](#).

Thermal and viscous contrast in quantum Hall scanning-probe images. P. Shubham Parashar.
Under review at *Physica E* (2026). [arXiv:2607.16315](#).

REFEREED PUBLICATIONS

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas. P. Shubham Parashar and M. M. Fogler.
Physical Review B **113**, 155443 (2026). [arXiv:2601.16402](#).

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene. Z. J. Krebs *et al.*, including P. Shubham Parashar.
Physical Review Letters **136**, 136301 (2026), Editors' Suggestion. [arXiv:2409.19468](#).

EDUCATION AND RESEARCH

2022–2026 **Ph.D., Physics**, University of California San Diego. Advisor: Michael M. Fogler.
2017–2022 **Researcher**, University of California San Diego. Research with Benjamin Grinstein and Congjun Wu.
2016–2017 **M.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
2012–2016 **B.S., Physics**, Indian Institute of Science, Bangalore.

SELECTED RESEARCH CAPABILITIES

- **Analytical kinetic transport theory.** Angular-harmonic expansions of the Boltzmann equation, higher-gradient and finite-wave-number response, and controlled hydrodynamic closures.
- **Symmetry-designed observables.** Drive and device symmetries that suppress leading dissipative channels locally and isolate higher kinetic harmonics as measurable signals.
- **Finite-resolution transport inference.** Spatial voltage, current, and dissipation maps, thermoelectric and Hall contrast, and finite-resolution scanning-probe inference.

METHODS

Boltzmann kinetic theory · electron hydrodynamics · semiclassical and quantum transport · linear response · asymptotics · spatial transport inference · numerical modeling · Python · Mathematica · LaTeX

DISTINCTIONS

Bhadra Fellowship (2024) · UCSD Scholar's Grant (2017) · KVPY Fellowship (2012–2017)